

# BSAVA API Architecture & Interface Specification

| Document Metadata   | Details                                             |
|---------------------|-----------------------------------------------------|
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| Author              | Timberyard Architecture & Engineering Team          |
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| Status              | Technical Baseline                                  |
| Target Architecture | Composable MACH Ecosystem                           |

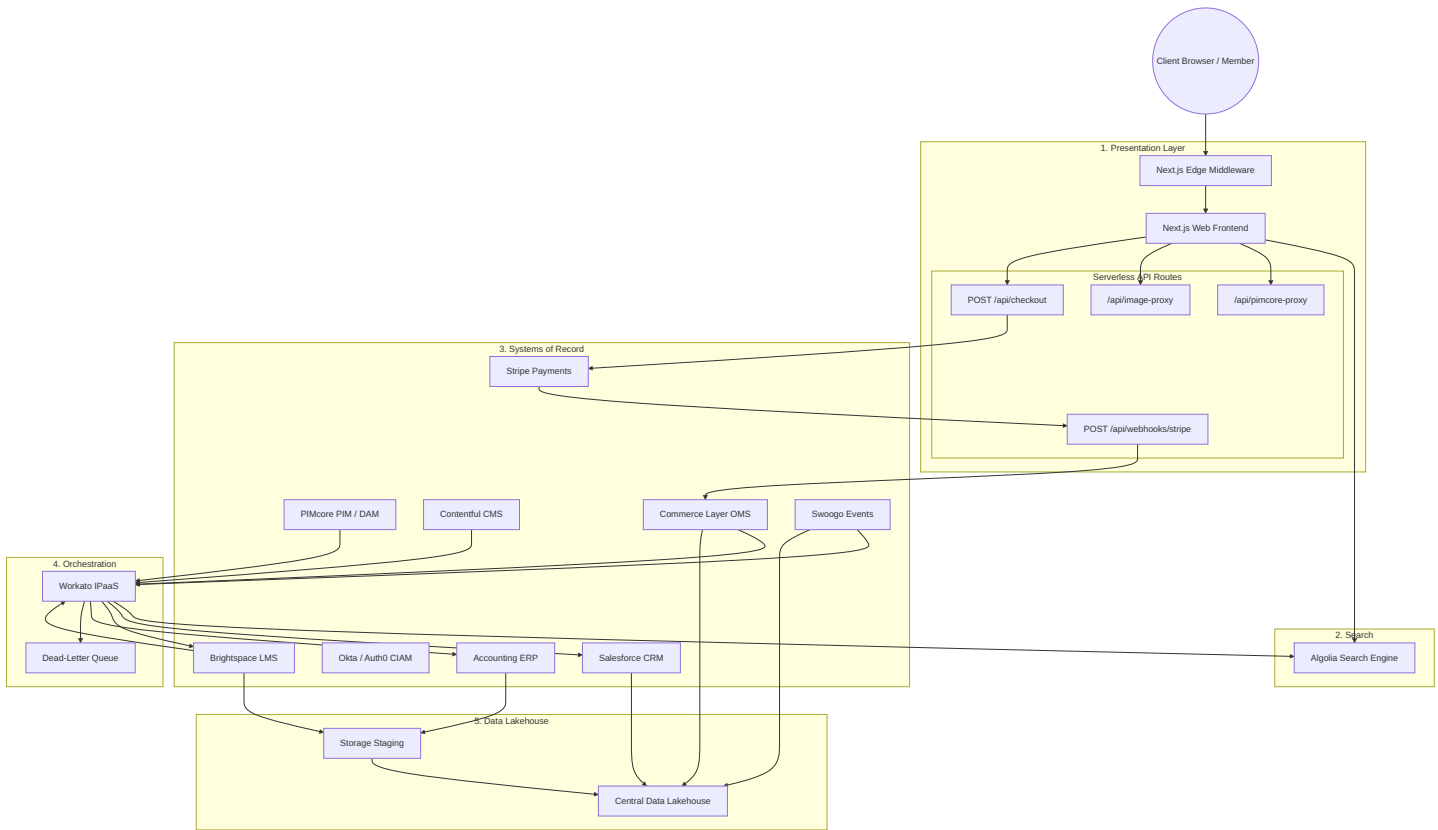
## 1. Purpose & Scope

This document catalogues every API endpoint, webhook, integration call, and data flow required by the BSAVA MACH platform, organised by **System of Record (SoR)**. It serves as the definitive interface contract for frontend engineers, backend integrators, and Workato recipe developers.

### Core Architectural Principles

- Single Source of Truth (SoR):** Every data domain is owned by exactly one operational platform.
- Token-Governed Orchestration:** Workato recipe executions adhere to a **1,000,000 task/year** quota via micro-batching and direct storage staging.
- Edge Security & Privacy:** Next.js Edge Middleware handles JWT authentication and CORS. All PII storage complies with UK GDPR in UK/EU cloud regions.

## 2. High-Level System Topology



## 3. API Specification by System of Record

### 3.1 Salesforce (CRM) — Contact & Member SoR

| #     | API / Endpoint                                                | Method         | Consumer              | Purpose                                                                                        |
|-------|---------------------------------------------------------------|----------------|-----------------------|------------------------------------------------------------------------------------------------|
| SF-01 | REST API —<br>/services/data/vXX.0/subjects/Contact           | GET /<br>PATCH | Workato               | Query and upsert member contact records                                                        |
| SF-02 | REST API —<br>/services/data/vXX.0/subjects/Account           | GET /<br>PATCH | Workato               | Query and upsert organisation accounts                                                         |
| SF-03 | REST API —<br>/services/data/vXX.0/subjects/Committee_Role__c | GET /<br>POST  | Workato               | Manage committee role assignments                                                              |
| SF-04 | REST API — /services/data/vXX.0/subjects/Order                | POST           | Workato               | Create order records from Commerce Layer webhooks                                              |
| SF-05 | REST API —<br>/services/data/vXX.0/query/?q=SELECT...         | GET            | Workato               | Hourly micro-batch query<br>( WHERE LastModifiedDate > {last_run} )<br>for lakehouse ingestion |
| SF-06 | Bulk API 2.0 — /services/data/vXX.0/jobs/ingest               | POST           | Workato               | Mass data load during historic migration<br>(Phase 3)                                          |
| SF-07 | SOAP API                                                      | POST           | Legacy                | Legacy integration support (migration<br>period only)                                          |
| SF-08 | CDC (ChangeEvent) — /data/ContactChangeEvent                  | Stream         | Workato<br>(optional) | Real-time change data capture (high token<br>cost — under review)                              |

**Authentication:** OAuth 2.0 (Connected App) with refresh token rotation.

**Outbound Webhooks from Salesforce:**

- CRM-to-CIAM profile sync push to Okta (< 5s latency) on Contact field changes
- Nightly batch reconciliation job for out-of-sync profiles

**Lakehouse Ingestion:** Hourly scheduled micro-batch → raw\_salesforce\_contacts (est. 8,760 tasks/year)

3.2 Okta / Auth0 (CIAM) — Identity & Session SoR

| #     | API / Endpoint                                    | Method           | Consumer           | Purpose                                                         |
|-------|---------------------------------------------------|------------------|--------------------|-----------------------------------------------------------------|
| OK-01 | Authentication API — /api/v1/authn                | POST             | Next.js Frontend   | User login, MFA challenge                                       |
| OK-02 | Token Endpoint — /oauth2/default/v1/token         | POST             | Next.js Frontend   | OAuth2 token exchange, refresh (15-min access / 14-day refresh) |
| OK-03 | Authorize Endpoint — /oauth2/default/v1/authorize | GET              | Next.js Frontend   | SSO redirect flow initiation                                    |
| OK-04 | Management API — /api/v1/users                    | GET / POST / PUT | Workato            | User CRUD, profile sync from Salesforce                         |
| OK-05 | System Log API — /api/v1/logs                     | GET              | Workato            | Security audit log streaming to Data Lakehouse                  |
| OK-06 | Backchannel Logout — /api/v1/sessions             | DELETE           | Salesforce / Admin | Global session revocation forcing re-authentication             |
| OK-07 | UserInfo Endpoint — /oauth2/default/v1/userinfo   | GET              | Next.js Frontend   | Fetch user profile claims                                       |

**Authentication:** API Token (Management API), OAuth 2.0 PKCE (Frontend flows).

**JWT Token Lifecycle:**

- Access tokens: 15-minute lifespan, carried as Bearer tokens
- Refresh tokens: httpOnly secure cookie, 14-day validity
- Edge Middleware validates JWT on every request; silent refresh on expiry

**Lakehouse Ingestion:** System Log API polled daily → security monitoring tables

3.3 PIMcore (PIM / DAM) — Product Catalog & Entitlements SoR

| #     | API / Endpoint                                                     | Method | Consumer         | Purpose                                                                                                                                  |
|-------|--------------------------------------------------------------------|--------|------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| PM-01 | GraphQL Data Hub — /pimcore-graphql-webservices/bsava?apikey={key} | POST   | Next.js Frontend | Multi-class product listing queries ( getBookListing , getEbookListing , getEventListing , getCourseListing , getMembershipTierListing ) |
| PM-02 | GraphQL Data Hub — getProduct(id)                                  | POST   | Next.js Frontend | Single product detail fetch with inline fragments ( Book , Ebook , Event , Course , MembershipTier )                                     |

| #     | API / Endpoint                                           | Method     | Consumer                                     | Purpose                                                           |
|-------|----------------------------------------------------------|------------|----------------------------------------------|-------------------------------------------------------------------|
| PM-03 | GraphQL Data Hub — legacy <code>getProductListing</code> | POST       | Next.js Frontend                             | Fallback query if multi-class schema unavailable                  |
| PM-04 | REST API — <code>/api/products</code>                    | POST / PUT | Workato                                      | Create/update event product records (Swoogo → PIMcore sync)       |
| PM-05 | REST API — <code>/api/products</code>                    | POST / PUT | Workato                                      | Create/update course product records (Brightspace → PIMcore sync) |
| PM-06 | REST API — <code>/api/entitlements</code>                | GET        | Workato / Commerce Layer                     | Query entitlement definitions and access rules                    |
| PM-07 | REST API — Asset endpoint                                | GET        | Next.js (via <code>/api/image-proxy</code> ) | Serve product images and digital assets                           |

**Authentication:** Bearer Token ( `PIMCORE_REST_API_TOKEN` ) for REST; API Key query parameter for GraphQL Data Hub.

**Outbound Webhooks from PIMcore:**

| Event                        | Destination       | Purpose                     |
|------------------------------|-------------------|-----------------------------|
| <code>product.created</code> | Workato → Algolia | Index new product in search |
| <code>product.updated</code> | Workato → Algolia | Update search index record  |
| <code>product.deleted</code> | Workato → Algolia | Remove from search index    |

**Lakehouse Ingestion:** Nightly cron at 01:00 UTC → `snap_pimcore_products` (365 tasks/year)

### 3.4 Contentful (CMS) — Editorial Content SoR

| #     | API / Endpoint                                                                              | Method     | Consumer         | Purpose                                                            |
|-------|---------------------------------------------------------------------------------------------|------------|------------------|--------------------------------------------------------------------|
| CF-01 | Content Delivery API (CDA) — <code>getEntries</code> ( <code>content_type: article</code> ) | GET        | Next.js Frontend | Fetch articles by slug, latest articles, all articles for indexing |
| CF-02 | CDA — <code>getEntries</code> ( <code>content_type: header</code> )                         | GET        | Next.js Frontend | Fetch site header navigation                                       |
| CF-03 | CDA — <code>getEntries</code> ( <code>content_type: footer</code> )                         | GET        | Next.js Frontend | Fetch site footer content                                          |
| CF-04 | CDA — <code>getEntries</code> ( <code>content_type: author</code> )                         | GET        | Next.js Frontend | Fetch author profiles and author-article associations              |
| CF-05 | Content Preview API — <code>preview.contentful.com</code>                                   | GET        | Next.js Frontend | Draft content preview                                              |
| CF-06 | Content Management API (CMA)                                                                | PUT / POST | Editorial team   | Content authoring (not consumed by frontend)                       |

**Authentication:** Delivery Token (CDA), Preview Token (Preview API), Management Token (CMA).

**Outbound Webhooks from Contentful:**

| Event                | Destination       | Purpose                                               |
|----------------------|-------------------|-------------------------------------------------------|
| <code>publish</code> | Workato → Algolia | Upsert article into <code>content</code> search index |

| Event     | Destination       | Purpose                                |
|-----------|-------------------|----------------------------------------|
| unpublish | Workato → Algolia | Remove stale article from search index |

**Lakehouse Ingestion:** Nightly cron at 01:00 UTC → snap\_contentful\_articles (365 tasks/year)

### 3.5 Commerce Layer (OMS) — Cart, Order & Membership Tier SoR

| #     | API / Endpoint                                    | Method        | Consumer         | Purpose                                           |
|-------|---------------------------------------------------|---------------|------------------|---------------------------------------------------|
| CL-01 | REST API — /api/orders                            | POST          | Next.js Frontend | Create order from cart                            |
| CL-02 | REST API — /api/line_items                        | POST / DELETE | Next.js Frontend | Add/remove cart items                             |
| CL-03 | REST API — /api/orders/{id}                       | PATCH         | Next.js Frontend | Update order (apply discounts, customer context)  |
| CL-04 | REST API — /api/customers                         | GET / POST    | Workato          | Customer record lookup/creation                   |
| CL-05 | REST API — /api/subscriptions                     | GET           | Workato          | Query membership subscription tier states         |
| CL-06 | REST API — /api/orders?filter[updated_at_gt]={ts} | GET           | Workato          | Hourly micro-batch order extraction for lakehouse |

**Authentication:** OAuth 2.0 Client Credentials (API credentials).

**Outbound Webhooks from Commerce Layer:**

| Event                | Destination           | Purpose                                    |
|----------------------|-----------------------|--------------------------------------------|
| order.placed         | Workato → Salesforce  | Create/update CRM contact and order record |
| order.placed         | Workato → Swoogo      | Confirm delegate registration status       |
| order.placed         | Workato → Brightspace | Enrol customer in purchased course         |
| order.placed         | Workato → ERP         | Generate invoice and record payment        |
| subscription.updated | Workato → Salesforce  | Sync membership tier changes               |

**Lakehouse Ingestion:** Hourly micro-batch → raw\_commerce\_orders + raw\_commercelayer\_subscriptions (est. 8,760 tasks/year)

### 3.6 Stripe — Payment Processing SoR

| #     | API / Endpoint                                        | Method | Consumer                     | Purpose                                                                           |
|-------|-------------------------------------------------------|--------|------------------------------|-----------------------------------------------------------------------------------|
| ST-01 | Checkout Sessions — stripe.checkout.sessions.create() | POST   | Next.js /api/checkout        | Create hosted checkout session with line items, bundle discounts, promotion codes |
| ST-02 | PaymentIntents API                                    | POST   | Commerce Layer               | Initialise payment from OMS order                                                 |
| ST-03 | Webhook Signature — stripe.webhooks.constructEvent()  | —      | Next.js /api/webhooks/stripe | Verify inbound webhook authenticity                                               |

**Authentication:** Secret Key (server-side), Publishable Key (client-side elements).

Inbound Webhooks handled by BSAVA:

| Event                      | Handler                      | Purpose                                  |
|----------------------------|------------------------------|------------------------------------------|
| checkout.session.completed | Next.js /api/webhooks/stripe | Trigger fulfilment logic, log payment    |
| payment_intent.succeeded   | Workato                      | Confirm payment for Commerce Layer order |
| charge.succeeded           | Workato                      | Financial reconciliation logging         |

**Lakehouse Ingestion:** Hourly micro-batch → raw\_stripe\_payments (combined with Commerce Layer recipe)

3.7 Swoogo — Events & Conferences SoR

| #     | API / Endpoint                   | Method | Consumer | Purpose                                                                 |
|-------|----------------------------------|--------|----------|-------------------------------------------------------------------------|
| SW-01 | REST API — Event query           | GET    | Workato  | Fetch event details and ticket types                                    |
| SW-02 | REST API — Registration update   | PUT    | Workato  | Mark registration as confirmed after payment                            |
| SW-03 | REST API — Delegate roster query | GET    | Workato  | Twice-daily scheduled poll for delegate registration arrays (lakehouse) |

**Authentication:** API Key / OAuth2 (vendor-specific).

Outbound Webhooks from Swoogo:

| Event                  | Destination          | Purpose                                                  |
|------------------------|----------------------|----------------------------------------------------------|
| event.created          | Workato → PIMcore    | Create event product record (MVE threshold)              |
| event.updated          | Workato → PIMcore    | Update event product metadata                            |
| registration.completed | Workato → Salesforce | Find/create contact, log attendance, assign entitlements |

Data Mapping (Swoogo → PIMcore):

| Swoogo Field          | PIMcore Field                 |
|-----------------------|-------------------------------|
| event_name            | name                          |
| start_date / end_date | eventStartDate / eventEndDate |
| location              | venue                         |
| ticket_types[].price  | pricing.basePrice             |
| capacity              | stockQuantity                 |

**Lakehouse Ingestion:** Twice-daily poll → raw\_swoogo\_registrations

3.8 Brightspace / D2L (LMS) — Learning & CPD SoR

| #     | API / Endpoint             | Method | Consumer | Purpose                                               |
|-------|----------------------------|--------|----------|-------------------------------------------------------|
| BS-01 | Enrolment API              | POST   | Workato  | Create course enrolment after product purchase        |
| BS-02 | Data Hub — Bulk CSV export | GET    | Workato  | Daily pull of course completions and CPD credit files |

| #     | API / Endpoint                | Method | Consumer         | Purpose                                                                 |
|-------|-------------------------------|--------|------------------|-------------------------------------------------------------------------|
| BS-03 | Course API — Enrolled courses | GET    | Next.js Frontend | Fetch member's enrolled courses and progress %                          |
| BS-04 | SSO Launcher                  | GET    | Next.js Frontend | Generate signed SSO URL for course player redirect                      |
| BS-05 | Course Metadata API           | GET    | Workato          | Fetch course title, description, syllabus, CPD credits for PIMcore sync |

**Authentication:** OAuth 2.0 / LTI (Learning Tools Interoperability) for SSO.

**Outbound Events from Brightspace (ENS):**

| Event            | Destination          | Purpose                                      |
|------------------|----------------------|----------------------------------------------|
| course.created   | Workato → Algolia    | Index new course in search                   |
| course.updated   | Workato → PIMcore    | Update course product metadata               |
| course.updated   | Workato → Algolia    | Update course search record                  |
| course.completed | Workato → Salesforce | Record CPD completion, trigger renewal flows |

**Lakehouse Ingestion:** Daily Data Hub CSV → S3/ADLS staging → raw\_brightspace\_completions (365 tasks/year)

### 3.9 Accounting ERP — Financial Ledger SoR

| #      | API / Endpoint                         | Method | Consumer | Purpose                                                     |
|--------|----------------------------------------|--------|----------|-------------------------------------------------------------|
| ERP-01 | REST API — Invoice creation            | POST   | Workato  | Create invoice and record payment from Commerce Layer order |
| ERP-02 | REST API — Journal entry               | POST   | Workato  | Post general ledger journal entries                         |
| ERP-03 | REST API / Scheduled export — GL query | GET    | Workato  | Extract GL entries and monthly trial balances for lakehouse |

**Authentication:** API Key / OAuth2 (vendor-specific).

**Lakehouse Ingestion:** Nightly scheduled export → S3/ADLS staging → raw\_erp\_gl\_entries

### 3.10 Algolia — Search Index (Operational Cache)

| #     | API / Endpoint                     | Method     | Consumer                       | Purpose                                                       |
|-------|------------------------------------|------------|--------------------------------|---------------------------------------------------------------|
| AL-01 | Search Client SDK — products index | GET        | Next.js Frontend (client-side) | Instant product and event search                              |
| AL-02 | Search Client SDK — content index  | GET        | Next.js Frontend (client-side) | Instant article and editorial search                          |
| AL-03 | Admin API — Upsert record          | POST / PUT | Workato                        | Add/update product, event, course, or article in search index |
| AL-04 | Admin API — Delete record          | DELETE     | Workato                        | Remove unpublished or deleted records from index              |

**Authentication:** Search API Key (client-side, read-only), Admin API Key (server-side, write).

Index Sources:

| Index             | Source System | Sync Trigger                            |
|-------------------|---------------|-----------------------------------------|
| products          | PIMcore       | Webhook on product create/update/delete |
| content           | Contentful    | Webhook on publish/unpublish            |
| content (courses) | Brightspace   | Webhook on course create/update         |

3.11 Data Lakehouse — Enterprise Analytics

| #     | API / Endpoint                                 | Method        | Consumer           | Purpose                                                |
|-------|------------------------------------------------|---------------|--------------------|--------------------------------------------------------|
| LH-01 | Bulk stage loading (Snowpipe / ADLS connector) | PUT           | Workato            | Land JSON/CSV/Parquet batches from operational systems |
| LH-02 | Direct storage copy (S3 / ADLS / OneLake)      | File transfer | Workato            | Bulk file landing for datasets > 10,000 records        |
| LH-03 | SQL endpoint (JDBC / ODBC)                     | SELECT        | Power BI / Tableau | BI queries against Silver/Gold zone tables             |

Target Tables:

| Table                           | Source         | Ingestion Pattern            |
|---------------------------------|----------------|------------------------------|
| raw_salesforce_contacts         | Salesforce     | Hourly micro-batch           |
| raw_commercelayer_subscriptions | Commerce Layer | Hourly micro-batch           |
| raw_commerce_orders             | Commerce Layer | Hourly micro-batch           |
| raw_stripe_payments             | Stripe         | Hourly micro-batch           |
| raw_swoogo_registrations        | Swoogo         | Twice-daily poll             |
| raw_brightspace_completions     | Brightspace    | Daily CSV via Data Hub       |
| raw_erp_gl_entries              | ERP            | Nightly export               |
| snap_pimcore_products           | PIMcore        | Nightly snapshot (01:00 UTC) |
| snap_contentful_articles        | Contentful     | Nightly snapshot (01:00 UTC) |

4. Next.js Frontend API Routes (Implemented)

| #     | Route                | Method | Purpose                                                             | Upstream System |
|-------|----------------------|--------|---------------------------------------------------------------------|-----------------|
| FE-01 | /api/checkout        | POST   | Create Stripe checkout session with line items and bundle discounts | Stripe          |
| FE-02 | /api/webhooks/stripe | POST   | Handle checkout.session.completed webhook, verify signature         | Stripe          |
| FE-03 | /api/pimcore-proxy   | POST   | Proxy GraphQL queries to PIMcore Data Hub                           | PIMcore         |
| FE-04 | /api/image-proxy     | GET    | Proxy PIMcore asset images to frontend                              | PIMcore         |



## 5. Workato Orchestration Recipes Summary

| #    | Recipe                    | Trigger                                     | Source → Destination         | Priority |
|------|---------------------------|---------------------------------------------|------------------------------|----------|
| W-01 | Event Sync                | event.created / event.updated webhook       | Swoogo → PIMcore             | P1       |
| W-02 | Order Provisioning        | order.placed webhook                        | Commerce Layer → Salesforce  | P1       |
| W-03 | Post-Registration Sync    | registration.completed webhook              | Swoogo → Salesforce          | P2       |
| W-04 | Registration Confirmation | order.placed webhook                        | Commerce Layer → Swoogo      | P2       |
| W-05 | Course Enrolment          | order.placed webhook (if courseId metadata) | Commerce Layer → Brightspace | P3       |
| W-06 | Product Index Sync        | Product webhook                             | PIMcore → Algolia            | P3       |
| W-07 | Content Index Sync        | publish / unpublish webhook                 | Contentful → Algolia         | P3       |
| W-08 | Course Metadata Sync      | course.updated / nightly job                | Brightspace → PIMcore        | P4       |
| W-09 | Course Search Sync        | course.created / course.updated webhook     | Brightspace → Algolia        | P4       |
| W-10 | CPD Completion Sync       | course.completed event                      | Brightspace → Salesforce     | P4       |

## Estimated Annual Token Budget

| Recipe Group                               | Pattern            | Est. Tasks/Year              |
|--------------------------------------------|--------------------|------------------------------|
| Member Profile & Subscription Ingestion    | Hourly micro-batch | ~8,760                       |
| Financial Transaction & Order Ingestion    | Hourly micro-batch | ~8,760                       |
| Event Delegate Analytics                   | Twice-daily poll   | ~730                         |
| Learning & CPD Credit Ingestion            | Daily bulk pull    | ~365                         |
| Product & Content Catalog Snapshots        | Nightly cron       | ~730                         |
| Operational webhook recipes (W-01 to W-10) | Event-driven       | Variable                     |
| Estimated Total                            |                    | ~20,000 + operational volume |

## 6. Error Handling & Resilience

| Pattern           | Detail                                                                         |
|-------------------|--------------------------------------------------------------------------------|
| Retry Policy      | Exponential backoff with jitter, max 3–5 retries over 30 minutes               |
| Dead-Letter Queue | Failed payloads written to S3/Azure Blob DLQ; #platform-ops Slack alert        |
| Idempotency       | All consumers enforce dedup via unique event IDs / transaction hashes          |
| Rate Limiting     | Public API routes: 100 req/min/IP. Downstream systems enforce API token quotas |
| API Versioning    | All custom service interfaces use semantic versioning ( /api/v1/... )          |

## 7. Authentication Summary

| System         | Auth Method                                       | Token/Key Type                                |
|----------------|---------------------------------------------------|-----------------------------------------------|
| Salesforce     | OAuth 2.0 Connected App                           | Access Token + Refresh Token                  |
| Okta / Auth0   | OAuth 2.0 PKCE (frontend), API Token (management) | JWT Access + httpOnly Refresh Cookie          |
| PIMcore        | Bearer Token (REST), API Key (GraphQL)            | PIMCORE_REST_API_TOKEN                        |
| Contentful     | API Tokens                                        | Delivery, Preview, Management tokens          |
| Commerce Layer | OAuth 2.0 Client Credentials                      | Client ID + Secret                            |
| Stripe         | API Keys                                          | Secret Key (server), Publishable Key (client) |
| Swoogo         | API Key / OAuth2                                  | Vendor-specific                               |
| Brightspace    | OAuth 2.0 / LTI                                   | SSO launcher tokens                           |
| Algolia        | API Keys                                          | Search Key (read), Admin Key (write)          |

End of API Architecture Document.